
EMBEDDED HACKING

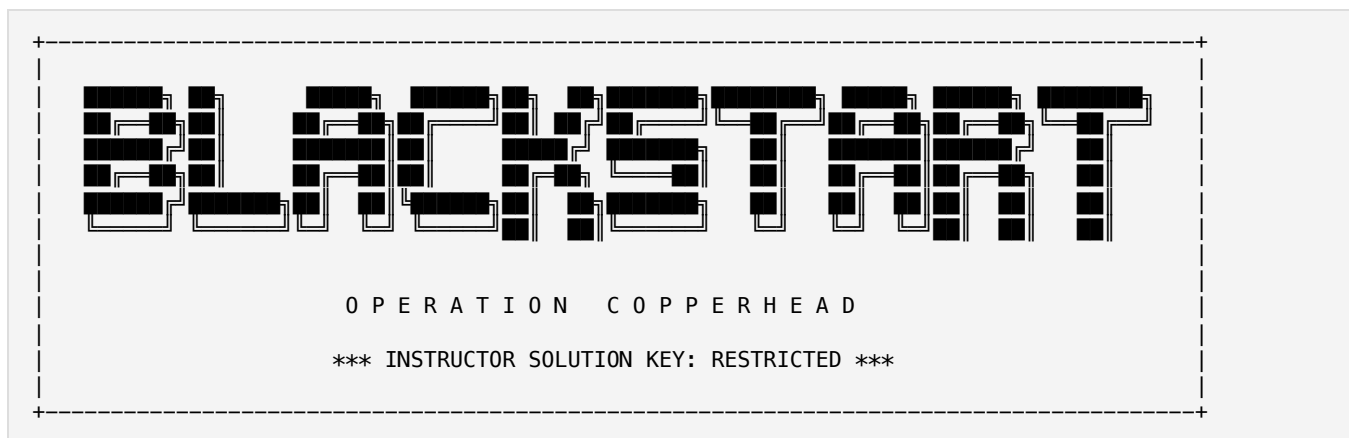
Reverse Engineering the Raspberry Pi Pico 2

Operation Copperhead - Instructor Solution Key

ARM Cortex-M33 • GDB • Ghidra

September 25, 2026

For educational and defensive security research only



The task and criterion headings in this key are word-for-word identical to CTF-02-R.md , so a student can match each criterion one-to-one.

Artifact Identity

Artifact	Value
Student image	CTF-02.bin
Flash image	CTF-02.uf2
Target	Raspberry Pi Pico 2 / RP2350 ARM Cortex-M33
Image base	0x10000000
Console	USB-CDC virtual COM, 115200 8N1

```
CTF-02.bin 85330C37CD0897746B1AF447E4BAC371DDE2042ABD2D61D58A61FE2A8EEF3537
CTF-02.uf2 F3CD4840260DB820D792758CECACC5297BEF1971B9EACF7601279256D8AF1EAB
```

Proof tool: `python3 scripts/verify_ctf.py` returns 26/26 checks passed against CTF-02.bin .

Task 1: Setup and Initial Analysis (12 points)

Solution

Criterion 1.1: Ghidra Project Setup (3 points). Import CTF-02.bin as Raw Binary , language ARM:LE:32:Cortex , base address 0x10000000 , then run auto-analysis.

Criterion 1.2: Vector Table Decoding (3 points). First 32 bytes of CTF-02.bin :

```
00 20 08 20 5B 01 00 10 1B 01 00 10 1D 01 00 10
11 01 00 10 11 01 00 10 11 01 00 10 11 01 00 10
```

Evidence	Answer
Vector table base	0x10000000
Initial SP	0x20082000
Reset pointer (as stored)	0x1000015B
Reset instruction address	0x1000015A

Criterion 1.3: main() and Status-Loop Addresses (4 points).

Element	Address
main()	0x100002E8
Recurring status loop start	0x1000034C
Loop back-edge (b.n 0x1000034C)	0x1000044E

Criterion 1.4: Thumb Addressing and Literal Pool (2 points). The stored reset pointer 0x1000015B has bit 0 set, selecting Thumb mode. Clearing bit 0 gives 0x1000015A. A representative literal pool entry is 0x100004B4, which holds 0x1000C4C8, the address of the format string "BLOCK STATE: %s", loaded by `ldr r0, [pc, #308]` at 0x1000037C.

Supporting Reference: SRAM Symbols (Ghidra names to semantic roles).

Ghidra Label	SRAM Address	Section	Role	Source Symbol
DAT_20001188	0x20001188	.data	Block length double	g_telemetry
DAT_20001198	0x20001198	.data	Signal key seed	g_auth_seed
DAT_2000119C	0x2000119C	.data	Track current	g_block_current
DAT_20001ECC	0x20001ECC	.bss	Dispatch state	g_dispatch_state
DAT_20001ED0	0x20001ED0	.bss	Fault poll counter	g_fault_polls
DAT_20001ED4	0x20001ED4	.bss	Input line buffer	g_linebuf
DAT_200020D4	0x200020D4	.bss	Input write index	g_lineidx
DAT_200020D8	0x200020D8	.bss	Operator state	g_operator_state
DAT_200020DC	0x200020DC	.bss	Derived signal key	g_signal_key

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 1.1: Ghidra Project Setup	3	Correct project name, ARM Cortex 32-bit little endian, base 0x10000000	One item off	Not set up
Criterion 1.2: Vector Table Decoding	3	Correct base, initial SP, reset pointer	One missing	Not found

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 1.3: main() and Status-Loop Addresses	4	Both addresses correct	One correct	Neither found
Criterion 1.4: Thumb Addressing and Literal Pool	2	Bit 0 cleared and one pool entry traced to its string	Partial	Incorrect

Instructor Notes & Assembly

- Confirm the Ghidra import used `Raw Binary`, `ARM:LE:32:Cortex`, base `0x10000000`, and that auto-analysis completed before any address was read.
- Verify `main()` is `0x100002E8`, the loop head is `0x1000034C`, and the back-edge is `0x1000044E`.
- For Criterion 1.4, accept any correctly traced literal pool entry; the pool entry `0x100004B4` holding `0x1000C4C8` (`"BLOCK STATE: %s"`, loaded at `0x1000037C`) is the reference example.
- The SRAM symbol table is supporting reference material, not a separate scored criterion.

Task 2: Find and Patch Bug #1: The Miscalibrated Release Threshold (15 points)

Solution

Criterion 2.1: Locate Compare Sites A and B (6 points).

```
10000302: 2b5e  cmp r3, #94  @ 0x5e
10000312: 2b5e  cmp r3, #94  @ 0x5e
```

Site	Address	File Offset	Original Bytes	Original Instruction
A	0x10000302	0x0302	5E 2B	cmp r3, #94
B	0x10000312	0x0312	5E 2B	cmp r3, #94

Criterion 2.2: Correct Immediate-Value Reasoning (4 points). The source constant is `SAFE_THRESHOLD = 95` and the test is `x < 95`. For an unsigned value, `x < 95` is exactly `x <= 94`, so the compiler emits `cmp r3, #94`. The correct limit is `60`, so the test is `x < 60`, which is `x <= 59`. The correct patched immediate is `0x3B (59)`, not `0x3C (60)`.

Criterion 2.3: Patch Compare Sites A and B (4 points).

Site	Address	File Offset	Original	Patched	After
A	0x10000302	0x0302	5E 2B	3B 2B	cmp r3, #59
B	0x10000312	0x0312	5E 2B	3B 2B	cmp r3, #59

Criterion 2.4: Explain Why Both Sites Must Be Patched (1 point). Site A drives the `BLOCK STATE` line and site B drives the `AUTO TRAIN` decision. Patching only site A makes the console read `CRITICAL` while the automated dispatch still says `AUTHORIZED`. Frozen reading `87 : 87 <= 94` is true (wrong); `87 <= 59` is false (correct).

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 2.1: Locate Compare Sites A and B	6	Both addresses and original bytes	One site	Not found
Criterion 2.2: Correct Immediate-Value Reasoning	4	Explains the <code><</code> to <code><=</code> transform and gives <code>0x3B</code>	Correct value, no reasoning	Wrong value
Criterion 2.3: Patch Compare Sites A and B	4	Both byte changes verified	One site	Not patched
Criterion 2.4: Explain Why Both Sites Must Be Patched	1	Clear explanation of the two independent comparisons	Vague	Missing

Instructor Notes & Assembly

- Both sites must be patched: `0x10000302` for `BLOCK STATE` and `0x10000312` for the `AUTO TRAIN` decision.
- The correct immediate is `0x3B` (59), not `0x3C` (60).
- Verify the byte changes on hardware; the corrected console reads `CRITICAL` and `HELD`.
- Ghidra ARM/Thumb Context Note:** In raw `.bin` files, patching an instruction that precedes an `IT` block (`ite ge`) using the GUI *Patch Instruction* action triggers Ghidra's `ReDisassembleCommand`. The re-disassembler encounters an internal context register conflict when trying to re-declare the `ITBlock` context over existing instructions, collapsing Thumb decoding into 32-bit ARM mode and swallowing Site B (`0x10000312`). Students must patch using the Bytes window workflow (Clear `C` -> edit byte `5E` -> `3B` in Bytes window with pencil icon -> Disassemble `D`) to keep Site B visible and cleanly aligned.

Task 3: Find and Patch Bug #2: The False TRACK Banner (10 points)

Solution

Criterion 3.1: Locate the Banner String (3 points).

String	Address
"TRACK: NORMAL\r"	<code>0x1000C4B8</code>
"NORMAL" substring to patch	<code>0x1000C4BF</code>

The string is loaded in `main` and printed once at boot; it never reads the sensor.

Criterion 3.2: Patch Six Characters (4 points). `NORMAL` and `DANGER` are both six ASCII characters, so the patch preserves the length.

Address Range	Original Bytes	Patched Bytes
<code>0x1000C4BF</code> - <code>0x1000C4C4</code>	<code>4E 4F 52 4D 41 4C</code>	<code>44 41 4E 47 45 52</code>

Criterion 3.3: Character-by-Character Documentation (2 points).

Address	Original Char	Original Byte	Patched Char	Patched Byte
0x1000C4BF	N	4E	D	44
0x1000C4C0	O	4F	A	41
0x1000C4C1	R	52	N	4E
0x1000C4C2	M	4D	G	47
0x1000C4C3	A	41	E	45
0x1000C4C4	L	4C	R	52

Criterion 3.4: Explain the Danger of a Hardcoded Status Word (1 point). The banner never consults the reading, so it reports a healthy track even while the frozen reading is dangerous, masking the hazard from the operator.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 3.1: Locate the Banner String	3	Correct address and cross-reference	Approximate	Not found
Criterion 3.2: Patch Six Characters	4	All six bytes changed, length preserved	Correct text, wrong bytes documented	Wrong length
Criterion 3.3: Character-by-Character Documentation	2	Original vs patched byte for all six characters	Partial	Missing
Criterion 3.4: Explain the Danger of a Hardcoded Status Word	1	Clear, specific reasoning	Generic	Missing

Instructor Notes & Assembly

- NORMAL and DANGER are both six characters; the patch must not change the string length or overwrite adjacent flash.
- Confirm the patch covers 0x1000C4BF through 0x1000C4C4 exactly.
- The banner is printed once at boot and never recomputed, so it is a separate defect from the threshold.

Task 4: Find and Patch Bug #3: The Block Length Constant (10 points)

Solution

Criterion 4.1: Locate the .data Double (3 points). The console prints BLOCK LENGTH: 3200 M from the format string at 0x1000C4F0. The value is a double in the .data init image at 0x1000EC60 (loaded into 0x20001188 at boot).

Criterion 4.2: IEEE-754 Bytes and Print Math (4 points). Eight bytes at 0x1000EC60 :

```

9A 99 99 99 99 99 09 40  -> 0x4009999999999999A  -> 3.2 km  -> 3200 m
7B 14 AE 47 E1 7A D4 3F  -> 0x3FD47AE147AE147B  -> 0.32 km -> 320 m

```

Criterion 4.3: Patch to Print 320 M (3 points).

File Offset Range	Flash Address Range	Original Bytes	Patched Bytes
0xEC60 - 0xEC67	0x1000EC60 - 0x1000EC67	9A 99 99 99 99 99 09 40	7B 14 AE 47 E1 7A D4 3F

At 0x1000EC68 the adjacent telemetry fields (03 00 00 00 flags and 07 00 crossing) are left untouched.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 4.1: Locate the .data Double	3	Correct address traced from the BLOCK LENGTH print	Approximate	Not found
Criterion 4.2: IEEE-754 Bytes and Print Math	4	Original and patched 8-byte double with print math (3200 M to 320 M)	Correct patch, no math	Wrong bytes
Criterion 4.3: Patch to Print 320 M	3	Console shows BLOCK LENGTH: 320 M	Wrong bytes	Not patched

Instructor Notes & Assembly

- The value is an 8-byte IEEE-754 double, not an integer. Follow the BLOCK LENGTH print into .data at 0x1000EC60 .
- The patched bytes 7B 14 AE 47 E1 7A D4 3F decode to 0.32 km (320 m).
- Confirm the adjacent telemetry fields at 0x1000EC68 (03 00 00 00 and 07 00) are left untouched.

Task 5: Find and Patch Bug #4: The Signal Seed (15 points)**Solution**

Criterion 5.1: Locate SIGNAL_SPEC and the Seed (5 points). The SIMPLE comparison in the loop is at 0x100003A6 :

```
100003a6: 4559    cmp r1, fp      ; fp = SIGNAL_SPEC = 0x2D879291 (pool 0x100004FC)
```

The corrupted seed is a 0x0A0A0A0A word in the .data init image at 0x1000EC70 (loaded into 0x20001198 at boot). The byte values 0A 0A 0A 0A also appear in the tbb jump table at 0x100003D4 ; those are not the seed.

Criterion 5.2: Patch the Seed (4 points). The required seed is the ChaCha expand word 0x6B206574 ("te k"), stored little-endian as 74 65 20 6B .

File Offset Range	Flash Address Range	Original Bytes	Patched Bytes
0xEC70 - 0xEC73	0x1000EC70 - 0x1000EC73	0A 0A 0A 0A	74 65 20 6B

Criterion 5.3: Explain the ARX Derivation (3 points). `derive_session_key(seed, iv)` works exactly like this:

1. Set `a = seed`, `b = iv`, `c = 0x61707865`, `d = 0x3320646E`.
2. Run four ChaCha quarter-rounds. A quarter-round runs four phases with rotate amounts 16, 12, 8, 7. Each phase is: `a = a + b`; `d = d XOR a`; `d = rotate_left(d, s)`; `c = c + d`; `b = b XOR c`; `b = rotate_left(b, s)`.
3. Return `a XOR d`.

The runtime IV is `derive_session_key(0x6B206574, 0) = 0x43C974F6`. The shipped seed `0x0A0A0A0A` gives `derive_session_key(0x0A0A0A0A, 0x43C974F6) = 0x915DCFF8` (MISMATCH). The honest seed `0x6B206574` gives `derive_session_key(0x6B206574, 0x43C974F6) = 0x2D879291`, which equals `SIGNAL_SPEC` and prints `OK`.

Criterion 5.4: Separate the Security Layers (3 points). The signal seed only controls the local `SIGNAL KEY` telemetry check. It does not authenticate the operator. The Argon2id plus XChaCha20-Poly1305 gate is a separate layer that requires the 12-word passphrase.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 5.1: Locate <code>SIGNAL_SPEC</code> and the Seed	5	<code>0x2D879291</code> located and wrong seed <code>0x0A0A0A0A</code> found	Partial	Not found
Criterion 5.2: Patch the Seed	4	Seed bytes changed to <code>74 65 20 6B</code>	Wrong byte	Not patched
Criterion 5.3: Explain the ARX Derivation	3	Correct trace of the per-cycle derivation	Vague	Missing
Criterion 5.4: Separate the Security Layers	3	Correctly explains what the seed fixes versus the gate	Generic	Missing

Instructor Notes & Assembly

- The seed is at `0x1000EC70` in the `.data` init image. The `0A 0A 0A 0A` bytes in the `tbb` jump table at `0x100003D4` are not the seed.
- The patched seed `74 65 20 6B` is the little-endian form of the ChaCha expand word `0x6B206574` ("tek").
- The seed only affects the local `SIGNAL KEY` check; the Argon2id plus XChaCha20-Poly1305 gate is a separate authentication layer.

Task 6: GDB Register Capture of the Derived Key (15 points)

Solution

Criterion 6.1: Breakpoint at the Derive Return (4 points). The per-cycle derive is `bl derive_session_key` at `0x10000352`. Break at the next instruction, `0x10000356`, and read `$r0`. On the shipped image it is `0x915DCFF8`.

```
(gdb) break *0x10000356
(gdb) continue
(gdb) print/x $r0      # 0x915DCFF8 on the corrupted image
```

Criterion 6.2: Inspect the Two Arguments (4 points). At the call entry `0x10000352` :

Register	Corrupted image	Meaning
<code>\$r0</code>	<code>0x0A0A0A0A</code>	Seed
<code>\$r1</code>	<code>0x43C974F6</code>	Derived IV

Criterion 6.3: Override the Register (4 points). With execution at `0x10000356` , set `$r0` to the spec key, then continue. The next cycle prints `SIGNAL KEY: 0x2D879291 OK` .

```
(gdb) set $r0 = 0x2D879291
(gdb) continue
```

Criterion 6.4: Watchpoint on the Stored Key (3 points). The key is stored in SRAM at `0x200020DC` (`str r0, [r6, #0]`).

```
(gdb) watch *0x200020DC
```

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 6.1: Breakpoint at the Derive Return	4	Correct address and <code>\$r0</code> read as the bug-derived key	Address off	Not found
Criterion 6.2: Inspect the Two Arguments	4	Live seed and derived IV captured at the second call	One correct	Missing
Criterion 6.3: Override the Register	4	<code>\$r0</code> set to <code>0x2D879291</code> and the next cycle shows <code>OK</code>	Partial	Missing
Criterion 6.4: Watchpoint on the Stored Key	3	Watchpoint on the SRAM key location documented	Approximate	Missing

Instructor Notes & Assembly

- Confirm the breakpoint is placed at `0x10000356` , the instruction after the `bl derive_session_key` at `0x10000352` .
- On the shipped image `$r0` reads `0x915DCFF8` ; `$r0` is the seed `0x0A0A0A0A` and `$r1` is the derived IV `0x43C974F6` .
- The stored key lives at `0x200020DC` ; accept the documented watchpoint.

Task 7: Recover the Ouroboros Authority Frame (10 points)

Solution

Criterion 7.1: Locate Salt, Nonce, Ciphertext, and Tag (4 points).

Component	Flash Address	Size
Ciphertext + Tag	0x1000CE94	64 B
Nonce	0x1000CED4	24 B
Salt	0x1000CEEC	16 B

Criterion 7.2: Document Argon2id Parameters and Payload Contract (2 points). Argon2id: memory 64 KiB, iterations 3, parallelism 1, output key 32 B, salt the 16 bytes above. XChaCha20-Poly1305 decrypts the 48-byte ciphertext with the 16-byte tag. The plaintext is 01 68 65 6C 6C 6F 0D 0A followed by zeros: byte 0 turns the GPIO 25 LED on, and bytes 1 through 7 are printed as hello plus carriage-return and newline.

Criterion 7.3: Authenticate with the 12-Word Passphrase (2 points). At the RESPONSE> prompt, type:

```
orbit olive ladder marble quartz canyon ripple saddle violet ember walnut falcon
```

Expected result (proven on hardware):

```
hello
AUTHORITY FRAME: VERIFIED
```

Criterion 7.4: State the Honest Quantum Boundary (2 points). Grover-style search halves the effective security exponent of a symmetric key, so a 256-bit key gives about 128 bits of quantum security. The construction uses classical symmetric and password-hashing primitives and does not implement NIST post-quantum standards.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 7.1: Locate Salt, Nonce, Ciphertext, and Tag	4	All three addresses correct in flash	Two correct	Not found
Criterion 7.2: Document Argon2id Parameters and Payload Contract	2	Correct memory/time/parallelism and payload layout	Partial	Missing
Criterion 7.3: Authenticate with the 12-Word Passphrase	2	AUTHORITY FRAME: VERIFIED, LED on, payload printed	Partial	Not shown
Criterion 7.4: State the Honest Quantum Boundary	2	Grover halves symmetric exponents; not strict PQC	Generic	Misstates

Instructor Notes & Assembly

- Verify the three component addresses in flash: ciphertext plus tag at `0x1000CE94` , nonce at `0x1000CED4` , salt at `0x1000CEEC` .
- The passphrase is fixed for the lab; a successful gate prints `hello` and `AUTHORITY FRAME: VERIFIED` and lights the on-board authority LED.
- Grade Criterion 7.4 on the honest boundary: 256-bit symmetric key maps to about 128 bits under Grover, and the design is not NIST post-quantum.

Task 8: Export and Verify (8 points)

Solution

Criterion 8.1: Export CTF-02_fixed.bin (1 point). Export the patched program from Ghidra (`File -> Export Program...` , `Binary Format`) as `CTF-02_fixed.bin` . The shipped image is 62,308 bytes.

Criterion 8.2: Convert to CTF-02_fixed.uf2 (1 point).

```
python uf2conv.py CTF-02_fixed.bin --base 0x10000000 --family 0xe48bff59 --output CTF-02_fixed.uf2
```

Criterion 8.3: Hardware Verification (4 points).

Before patching:

```
DEEPLINE METRO AUTHORITY
ADAPTIVE SIGNAL WINDOW: 38 MINUTES
USB-CDC 115200 8N1 | AUTHORIZED LAB CONSOLE
TRACK: NORMAL
BLOCK STATE: STABLE
AUTO TRAIN: AUTHORIZED
BLOCK LENGTH: 3200 M
FAULT POLLS: 1
SIGNAL KEY: 0x915DCFF8 MISMATCH
RESPONSE>
```

After all four patches:

```
DEEPLINE METRO AUTHORITY
ADAPTIVE SIGNAL WINDOW: 38 MINUTES
USB-CDC 115200 8N1 | AUTHORIZED LAB CONSOLE
TRACK: DANGER
BLOCK STATE: CRITICAL
AUTO TRAIN: HELD
BLOCK LENGTH: 320 M
FAULT POLLS: 1
SIGNAL KEY: 0x2D879291 OK
RESPONSE>
```

Criterion 8.4: Summary Table of All Patches (2 points).

#	Bug	File Offset	Flash Address	Original Bytes	Patched Bytes
1a	Operator threshold	<code>0x0302</code>	<code>0x10000302</code>	<code>5E 2B</code>	<code>3B 2B</code>

#	Bug	File Offset	Flash Address	Original Bytes	Patched Bytes
1b	Dispatch threshold	0x0312	0x10000312	5E 2B	3B 2B
2	TRACK banner	0xC4BF - 0xC4C4	0x1000C4BF - 0x1000C4C4	4E 4F 52 4D 41 4C	44 41 4E 47 45 52
3	Block length	0xEC60 - 0xEC67	0x1000EC60 - 0x1000EC67	9A 99 99 99 99 99 09 40	7B 14 AE 47 E1 7A D4 3F
4	Signal seed	0xEC70 - 0xEC73	0x1000EC70 - 0x1000EC73	0A 0A 0A 0A	74 65 20 6B

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 8.1: Export CTF-02_fixed.bin	1	Valid patched binary	Corrupted	Not submitted
Criterion 8.2: Convert to CTF-02_fixed.uf2	1	Correct base and family flags	Wrong flags	Not submitted
Criterion 8.3: Hardware Verification	4	Corrected console output confirmed on hardware	Some lines corrected	No verification
Criterion 8.4: Summary Table of All Patches	2	Complete address and before/after table	Missing entries	No table

Instructor Notes & Assembly

- Verify the exported image with `python3 scripts/verify_ctf.py`; the shipped check expects 26/26 checks passed against CTF-02.bin.
- Confirm the UF2 conversion used base `0x10000000` and family `0xe48bff59`.
- Serial Terminal Timing:** Note that `print_identity()` (TRACK: NORMAL) fires within the first 5 milliseconds of boot. In normal lab usage, PuTTY attaches after boot and will display the continuous 2-second status loop (BLOCK STATE: CRITICAL , AUTO TRAIN: HELD). To see the corrected banner, the student must pulse RUN to GND while PuTTY is open, or demonstrate the string change at `0x1000C4BF` via Ghidra static analysis.
- The shipped image is 62,308 bytes; confirm the exported corrected image is a valid patched binary with all four fixes present.

Task 9: Written Reflection (5 points)

Solution

Criterion 9.1: "Rushed Build" Is Not an Excuse (2 points). The rebuild shipped four constants that were never checked against their documented limits, which is exactly what produced the false-safe reading. Pressure explains why the checks were skipped, not why they should be skipped.

Criterion 9.2: One Engineering Practice per Failure Area (3 points).

- Physical limits (Bugs #1 and #3): one shared configuration header plus a build-time assertion that each compiled limit matches its documented value.
- Banner (Bug #2): remove static banners; a hardware-in-the-loop test that compares displayed state to the live register.
- Seed integrity (Bug #4): reproducible builds with golden artifact hash comparison so keys and seeds match the certified specification.
- Image authenticity: enable RP2350 hardware secure boot with OTP hash verification so a modified image will not run.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 9.1: "Rushed Build" Is Not an Excuse	2	Specific, grounded reasoning	Generic	Missing
Criterion 9.2: One Engineering Practice per Failure Area	3	Concrete practices for the bugs and for image authenticity	Names some	Missing

Instructor Notes & Assembly

- Grade the specificity of the reasoning, not the length of the prose.
- Require concrete practices across the failure areas, including at least one practice for image authenticity.

How To Breadboard

- **Raspberry Pi Pico 2** powered over USB.
- **USB-CDC virtual serial console:** open the Pico's COM port at 115200 baud, 8 data bits, no parity, 1 stop bit.
- **SWD debug probe:** connect SWCLK, SWDIO, GND, and 3.3 V to the Pico debug header for GDB inspection and register capture.
- No other peripherals are required; the authority LED is on-board.

Complete Grading Summary

Task	Title	Points
Task 1	Setup and Initial Analysis	12
Task 2	Find and Patch Bug #1: The Miscalibrated Release Threshold	15
Task 3	Find and Patch Bug #2: The False TRACK Banner	10
Task 4	Find and Patch Bug #3: The Block Length Constant	10
Task 5	Find and Patch Bug #4: The Signal Seed	15
Task 6	GDB Register Capture of the Derived Key	15

Task	Title	Points
Task 7	Recover the Ouroboros Authority Frame	10
Task 8	Export and Verify	8
Task 9	Written Reflection	5
TOTAL		100

Instructor Notes

Safety: Use only the supplied Pico 2, SWD probe, and firmware. Never connect the exercise to an operational railway, metro system, public network, military system, or third-party device.

Common Student Mistakes

- Patching only one threshold site (`0x10000302` or `0x10000312`), leaving one status line lying.
- Assuming the immediate equals the limit, producing an off-by-one boundary; the correct byte is `0x3B` (59), not `0x3C` (60).
- Replacing the banner string with a different length, corrupting adjacent flash; `NORMAL` and `DANGER` are both 6 bytes.
- Treating the block length as an integer and missing the 8-byte double in `.data`.
- Using the wrong `0.32` bytes and printing `316 M` instead of `320 M`.
- Treating the odd vector address `0x1000015B` as invalid instead of clearing bit 0 to get `0x1000015A`.
- Starting the seed patch at the wrong offset; the seed is at `0x1000EC70`.

Partial Credit Guidelines

- Award partial credit for one correct threshold site out of two, or for a correct immediate value without the `<` to `<=` reasoning.
- Award partial credit for a correct banner text with incorrectly documented bytes, or for partial character-by-character documentation.
- Award partial credit for a correct block-length patch without the IEEE-754 print math.
- Award partial credit for one of the two GDB argument captures, or for a partial register override.
- Award no credit for patches that change string length or overwrite adjacent flash.

Appendix: Expected Binary Diff

#	Bug	File Offset(s)	Flash Address(es)	Original Bytes	Patched Bytes
1a	Operator threshold	<code>0x0302</code>	<code>0x10000302</code>	<code>5E 2B</code>	<code>3B 2B</code>
1b	Dispatch threshold	<code>0x0312</code>	<code>0x10000312</code>	<code>5E 2B</code>	<code>3B 2B</code>
2	TRACK banner	<code>0xC4BF - 0xC4C4</code>	<code>0x1000C4BF - 0x1000C4C4</code>	<code>4E 4F 52 4D 41 4C</code>	<code>44 41 4E 47 45 52</code>

#	Bug	File Offset(s)	Flash Address(es)	Original Bytes	Patched Bytes
3	Block length	0xEC60 - 0xEC67	0x1000EC60 - 0x1000EC67	9A 99 99 99 99 99 09 40	7B 14 AE 47 E1 7A D4 3F
4	Signal seed	0xEC70 - 0xEC73	0x1000EC70 - 0x1000EC73	0A 0A 0A 0A	74 65 20 6B

Four defects, five changed regions: two immediate bytes (0x3B 2B at each threshold), six banner bytes, eight block-length bytes, and four seed bytes.